

Low-Cost Humanoid Teleoperation: Wearable IMU-Based Human Pose Retargeting and Motion Tracking for a Simulated Humanoid Robot

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This project develops a low-cost lower-body humanoid teleoperation pipeline using wearable IMUs, ESP32-S3 wireless nodes, human pose reconstruction, retargeting, and simulation-based motion tracking. Instead of relying on expensive motion capture systems, our system streams fused IMU quaternions over UDP at 100 Hz, achieving end-to-end latency under 50 ms from wearable sensing to simulated humanoid output. The pipeline reconstructs a lower-body human skeleton from five physical IMUs, exports standardized pose clips, retargets those poses to a custom humanoid robot model, and trains a PPO-based motion-tracking policy with residual PD control in IsaacLab. Real robot deployment and full-body teleoperation remain future work, but the completed simulation pipeline demonstrates a practical, low-cost foundation for humanoid teleoperation at approximately \$180 in hardware cost.

1 Introduction

Humanoid robots are useful research platforms for studying locomotion, whole-body control, human-robot interaction, and embodied intelligence. However, controlling a humanoid robot is difficult because the robot has many degrees of freedom, unstable dynamics, contact-rich motion, and strict physical limits. One natural way to make humanoid control more intuitive is teleoperation: a human moves their own body, and the robot attempts to reproduce the same motion. This is especially appealing for lower-body control because humans can naturally demonstrate motions such as stepping, walking, turning, and shifting weight.

The main barrier is sensing. High-quality humanoid teleoperation often depends on optical motion capture systems, full-body virtual reality trackers, or other specialized tracking hardware. These systems can be accurate, but they are expensive, environment-dependent, and difficult for student robotics teams to deploy. Optical systems require a calibrated camera volume and can suffer from occlusion. RGB-D and markerless vision systems are easier to deploy, but the legs often occlude each other during walking. Wearable inertial measurement units (IMUs) provide a lower-cost alternative because they are self-contained, portable, and do not require an instrumented room.

This project investigates whether a sparse wearable IMU setup can provide enough lower-body motion information to drive a humanoid robot in simulation. The system uses BNO085 IMUs paired with ESP32-S3 microcontrollers. Each wearable node streams fused orientation quaternions over Wi-Fi/UDP to a compute node. The compute node filters and calibrates the orientation streams, aligns them to body segments, reconstructs a lower-body human skeleton, and exports the result as a standardized human pose representation. That human pose is then retargeted to a custom humanoid robot model, `ch_robot`, and evaluated in simulation with MuJoCo and IsaacLab.

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The project originally aimed to close the entire loop from wearable sensing to real humanoid deployment. During the quarter, we revised the final MVP toward simulation-based teleoperation and Sim2Sim validation. This was a deliberate safety and feasibility decision. Real robot deployment requires robust root motion estimation, contact-aware retargeting, stable learned control, and careful validation before commands can be sent to hardware. By the end of the quarter, we completed the core pipeline from wearable sensing to lower-body pose reconstruction, human-to-robot retargeting, simulation playback, and motion-tracking policy validation.

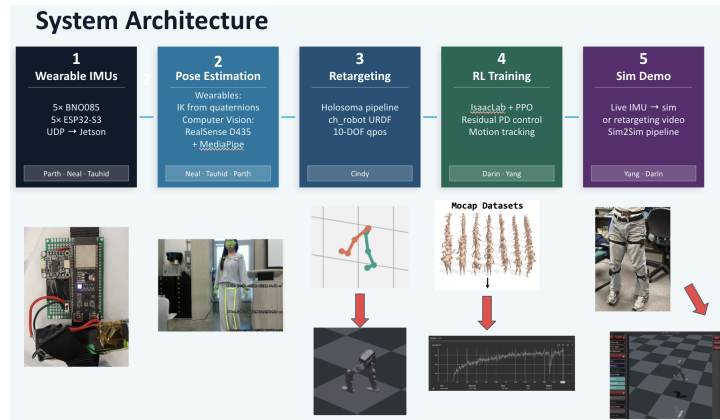


Fig. 1. Overview of the low-cost humanoid teleoperation pipeline. Wearable IMUs stream orientation data to the pose reconstruction layer, which produces a lower-body human pose for retargeting and simulated humanoid control.

The main contributions of this project are:

- A low-cost wearable inertial sensing platform using BNO085 IMUs and ESP32-S3 microcontrollers for real-time lower-body orientation capture.
- A real-time orientation-processing pipeline that validates, filters, smooths, calibrates, and aligns sparse IMU streams to human body segments.
- A lower-body kinematic reconstruction method that produces the nine human keypoints required for robot retargeting from as few as five physical IMUs.
- A human-to-robot retargeting pipeline that maps the reconstructed human lower-body pose to the custom `ch_robot` humanoid model.
- A simulation-based validation stack using IsaacLab, PPO-based motion tracking, residual PD control, and MuJoCo replay assets.

2 Related Works

2.1 Inertial Motion Capture and Wearable Sensing

Commercial inertial motion capture systems, such as Xsens MVN, use multiple body-mounted IMUs and biomechanical models to estimate full-body motion without external cameras [1]. These systems show that inertial sensing can capture human motion outside a motion capture studio, but they are still expensive and proprietary. Earlier hybrid systems combined inertial sensing with acoustic or ultrasonic ranging to reduce positional drift [2]. Our system follows the same broad idea of using orientations on human body segments, but it targets a lower-cost student-buildable platform using commodity BNO085 IMUs and ESP32-S3 microcontrollers.

2.2 Sparse-IMU Human Pose Estimation

Recent work has shown that plausible full-body pose can be reconstructed from a sparse set of IMUs. Sparse Inertial Poser fits a body model to sparse IMU measurements offline [3], Deep Inertial Poser uses a recurrent neural network for real-time pose reconstruction [4], and TransPose estimates both pose and global translation from six IMUs [5]. These systems motivate our sparse lower-body design: a small number of well-placed inertial sensors can contain enough information for useful pose reconstruction. Our approach differs by using an explicit orientation-based kinematic model rather than a learned full-body pose prior. This makes the system easier to debug and more appropriate for a small dataset and a short project timeline.

2.3 Orientation Estimation and Calibration

Estimating orientation from inertial sensors is a well-studied filtering problem. Mahony’s nonlinear complementary filter [6], Madgwick’s gradient-descent orientation filter [7], and Kalman-style inertial/magnetic fusion methods [8, 9] are common approaches for converting raw accelerometer, gyroscope, and magnetometer readings into orientation estimates. In our system, the BNO085 performs this fusion on-chip and directly outputs a rotation-vector quaternion. This moves the hardest part of inertial orientation estimation off the host computer and lets the rest of the system focus on calibration and kinematics.

A separate issue is sensor-to-segment calibration. Since an IMU is strapped to a body segment rather than rigidly attached to bone, the sensor frame is not automatically aligned with the anatomical segment frame. Prior gait-analysis work computes joint angles from relative IMU orientations and estimates segment alignment through calibration or functional motions [10, 11]. Our system uses a neutral-pose calibration and a documented sensor mounting convention to remove most mounting offsets.

2.4 Human-to-Robot Retargeting

Human-to-robot retargeting converts human motion into robot joint commands while preserving the visible intent of the motion. This is challenging because humans and robots have different limb lengths, joint axes, joint limits, and contact constraints. Whole-body retargeting systems often solve an optimization problem that balances human pose matching with robot feasibility [12, 13]. Our project uses the Holosoma retargeting pipeline as a starting point and adapts it to the custom `ch_robot` model. The lower body is especially difficult because foot contact, knee bend, and root motion must remain physically plausible.

2.5 Motion Tracking with Reinforcement Learning

Physics-based character control and humanoid motion tracking are often trained with reinforcement learning. DeepMimic demonstrated that policies can imitate reference motion clips in simulation [14]. PPO is commonly used for continuous-control robotics because it is stable and relatively simple to tune [15]. Later work such as AMP uses learned motion priors to produce naturalistic character control [16]. Our project follows this general direction, but the intended online reference is not clean motion capture data. It is noisy, partial, real-time IMU-derived motion. For this reason, our simulation stack uses residual PD control, where retargeted joint targets provide the nominal motion and the policy learns stabilizing corrections.

2.6 Simulation-Based Validation

Simulation is a critical step before real humanoid deployment. MuJoCo provides a widely used physics engine for robot modeling and control [17], while IsaacLab provides a modern reinforcement learning framework for robot training [18]. We use simulation to validate retargeted motion and learned tracking behavior before attempting any real robot deployment. This Sim2Sim stage is important because a motion or control policy that fails across simulators is unlikely to be safe on hardware.

3 Technical Material

3.1 System Architecture

The system has four main technical layers: wearable inertial sensing, lower-body pose reconstruction, human-to-robot retargeting, and simulated motion tracking. The wearable layer produces real-time orientation streams from body-mounted IMUs. The pose reconstruction layer converts those orientations into a lower-body human skeleton. The retargeting layer maps the human skeleton into `ch_robot` generalized coordinates. The simulation and control layer evaluates whether the retargeted motion can drive a humanoid model in physics simulation.

This organization is important because the wearable system does not directly output robot commands. The IMUs produce segment orientations. A kinematic reconstruction layer turns those orientations into human joint positions and joint rotations. The retargeting layer then converts the human representation into robot `qpos` and `qvel`. This separation makes the system easier to debug: errors in sensing, human pose reconstruction, robot retargeting, and control can be inspected independently.

3.2 Wearable Inertial Sensing

Each wearable node consists of a BNO085 IMU and an ESP32-S3 microcontroller. The BNO085 is not just a raw accelerometer and gyroscope; it contains an onboard sensor-fusion processor that outputs a fused absolute orientation quaternion. This design choice simplified the host-side software because the receiver did not need to integrate raw inertial measurements. Instead, it could start from orientation quaternions and focus on packet validation, calibration, sensor-to-segment alignment, and kinematic reconstruction.

The ESP32-S3 streams orientation data over Wi-Fi using UDP. UDP was chosen because real-time teleoperation values freshness over guaranteed delivery. A new orientation packet supersedes the previous one, so retransmitting stale data is less useful than dropping it. The rotation-vector report is streamed at 100 Hz, which gives a 10 ms nominal sample interval. The full intended lower-body setup uses seven nodes: pelvis, left thigh, left shank, left foot, right thigh, right shank, and right foot. The final MVP used five nodes: pelvis, both thighs, and both shanks.

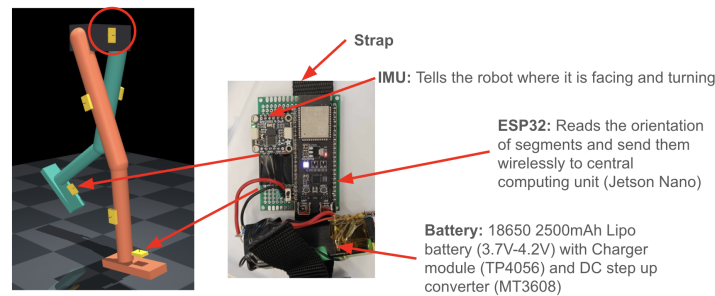


Fig. 2. Annotated wearable sensing node used in the project. Each node combines a BNO085 IMU, ESP32-S3 wireless microcontroller, battery, and power regulation modules for portable lower-body motion capture.

Each packet is a compact 40-byte binary message containing the physical board ID, assigned body segment ID, sequence number, timestamp, fused quaternion, accuracy estimate, and sensor status. The distinction between board identity and segment identity is important. The `sensor_id` identifies the physical ESP32 board, while the `segment_id` identifies the body placement for the current session. This allows boards to be swapped or reused without changing the body-segment organization of the pose pipeline.

The packet bandwidth is small. At approximately 40 bytes per packet and 100 Hz, each node transmits about 3.9 KB/s. For the five-node MVP, this is about 19.5 KB/s, which is trivial for Wi-Fi. The main practical issues

were not bandwidth but stability, latency, and jitter. Early testing showed that Wi-Fi power saving could produce bursty packet delivery, so the firmware disabled Wi-Fi sleep and increased transmit power to make the stream steadier. The final MVP achieved sub-50 ms end-to-end latency from wearable sensing to host-side pose output. End-to-end latency was estimated by halving the UDP round-trip time measured via a ping-pong script (100 packets at 50 ms intervals per node); processing overhead (filtering and IK) is bounded by the 10 ms loop period at 100 Hz.

Each node is self-contained and battery powered. The wearable power stack uses an 18650 lithium cell, a charge/protection module, a boost converter, and an inline switch. This made the nodes portable enough for lower-body experiments and avoided tethering the user to a bench power supply.

3.3 Orientation Processing and Calibration

The orientation-processing layer converts raw incoming orientation packets into calibrated body-segment orientations. The receiver first stores the newest valid packet for each body segment. Invalid packets are rejected using several checks: quaternion norm validation, sequence-number checks for stale or out-of-order packets, and spike rejection for implausible angular jumps. Accepted orientations are smoothed with spherical linear interpolation (SLERP), which reduces jitter while preserving valid rotational motion [19].

Before recording or retargeting, the user performs a neutral-pose calibration. During calibration, the subject stands still while the system collects multiple orientation samples for each required segment. These samples are averaged in quaternion space after aligning signs to the same hemisphere, preventing equivalent positive and negative quaternion representations from canceling during averaging [20]. The neutral pose becomes the reference orientation for the session. Later live orientations are expressed relative to this neutral reference:

$$q_{\text{calibrated}} = q_{\text{neutral}}^{-1} q_{\text{live}}.$$

This makes the neutral standing pose map to identity and reduces the effect of small mounting differences across sessions.

After neutral calibration, the system removes the fixed mounting rotation between the sensor frame and the body segment frame. The body/world convention used in the project is +X forward, +Y left, and +Z up. The sensor mounting convention assumes that the sensor's outward axis points away from the body segment, while another known axis points up the limb or toward the toes depending on placement. Given this sensor-to-segment mounting transform, the body-segment orientation is computed as:

$$R_{\text{world} \leftarrow \text{segment}} = R_{\text{world} \leftarrow \text{sensor}} R_{\text{sensor} \leftarrow \text{segment}}^{-1}.$$

This step is necessary because an IMU reports the orientation of the device case, not the anatomical orientation of the thigh, shank, foot, or pelvis.

3.4 Lower-Body Pose Reconstruction

Once each available body segment has a calibrated world orientation, the system reconstructs a lower-body skeleton through orientation-based forward kinematics. A joint rotation is defined as the child segment expressed in the parent segment's frame:

$$R_{\text{joint}} = R_{\text{parent}}^{-1} R_{\text{child}}.$$

Using this rule, the hip rotation is pelvis-to-thigh, the knee rotation is thigh-to-shank, and the ankle rotation is shank-to-foot.

The 3D joint positions are then recovered from a fixed lower-body model. Starting at the pelvis, the system places the hip centers using pelvis orientation and hip spacing. It then walks down each leg by rotating fixed

bone-length vectors through the corresponding segment orientations:

$$p_{\text{knee}} = p_{\text{hip}} + R_{\text{thigh}}[0, 0, -\ell_{\text{thigh}}]^T,$$

$$p_{\text{ankle}} = p_{\text{knee}} + R_{\text{shank}}[0, 0, -\ell_{\text{shank}}]^T,$$

$$p_{\text{toe}} = p_{\text{ankle}} + R_{\text{foot}}[\ell_{\text{toe}}, 0, 0]^T.$$

The default body dimensions include hip spacing, thigh length, shank length, foot length, heel length, and toe length. Since orientation quaternions do not determine bone lengths, these dimensions must come from a body model or manual measurement. The exact lengths affect the scale of the reconstructed positions, but the relative joint rotations come from the IMU orientations.

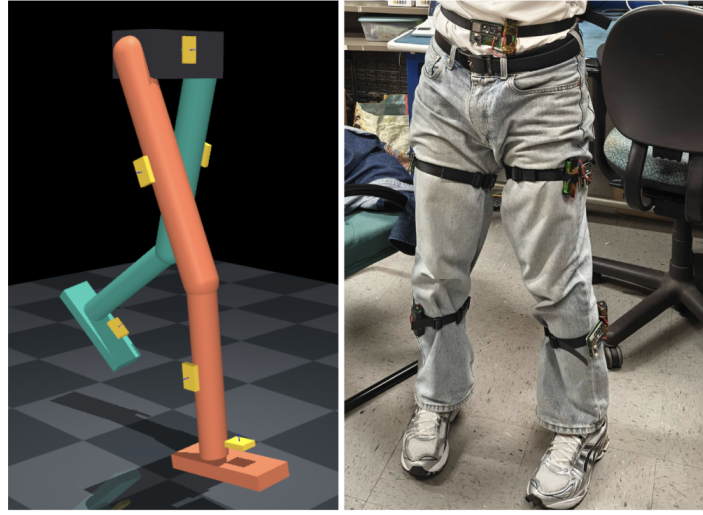


Fig. 3. IMU placement configurations used by the lower-body pose pipeline. The final MVP used the five-sensor shanks configuration, while the full intended system used seven sensors including both feet.

A central design feature is sparse-sensor reconstruction. The robot retargeting interface expects nine lower-body keypoints: pelvis/root, left and right hips, knees, ankles, and toe-base points. The five-IMU MVP does not directly measure foot orientations, and it does not mount sensors on the toes. To close this gap, the system uses two graceful-degradation rules. First, missing distal segment orientations inherit the proximal parent's orientation. For example, if a foot IMU is missing, the foot orientation is estimated from the corresponding shank. Second, toe-base points are generated from the lower-body model rather than directly measured. This allows five physical IMUs to produce the full nine-keypoint representation needed by the retargeting system.

3.5 Root Motion and Walking

A major limitation of orientation-only IMU sensing is that it does not directly measure global pelvis translation. If the pelvis is fixed in place, the system can reconstruct posture but not true walking displacement. This is sufficient for non-walking lower-body motions and debugging, but it is not enough for full locomotion teleoperation.

To address this, the system includes a free-root mode inspired by zero-velocity update methods used in pedestrian tracking [21]. The key physical assumption is that, during walking, the stance foot is momentarily

stationary with respect to the ground. The system selects a stance foot, pins it to a world anchor, and solves the pelvis position relative to that anchor:

$$p_{\text{pelvis}} = p_{\text{anchor}} - \Delta p_{\text{stance foot}}.$$

When the swing foot plants, the anchor can switch to the new stance foot. This lets the reconstructed pelvis move through the world without directly integrating acceleration.

This method is useful, but it is not a complete replacement for motion capture. Contact is inferred from the kinematic model, so errors in foot orientation or foot placement can produce root errors. Yaw drift can accumulate over time, especially indoors where magnetometer readings may be disturbed. The method also works best with the full seven-IMU setup because foot orientation is directly measured. For the final report and demonstration, we therefore treat free-root tracking as a useful extension rather than a fully solved walking localization system.

3.6 Human Pose Handoff and Retargeting

The embedded-to-ML interface exports a nine-point lower-body pose rather than raw IMU packets. The required points are:

Spine1, LeftUpLeg, LeftLeg, LeftFoot, LeftToeBase,
RightUpLeg, RightLeg, RightFoot, RightToeBase.

Spine1 is the pelvis/root reference point. It is not a physical joint in the same anatomical sense as the hip, knee, or ankle. The toe-base points are generated from the foot model rather than directly measured.

The preferred recorded format is one .npz file per movement trial. The core arrays are joint_names, joint_pos_origin, timestamps_s, and fps. Additional metadata such as joint_pos_w, root_quat_wxyz, config, and required_segments is included for debugging and retargeting. The array joint_pos_origin stores positions relative to the frame-zero pelvis projected onto the floor, making clips easier to compare across trials.

The retargeting stage converts this human pose into robot generalized coordinates. The target robot is the custom ch_robot humanoid model. Its retargeted qpos trajectory has shape $T \times 17$, consisting of a floating base plus 10 lower-body joint coordinates. We adapted Holosoma to ch_robot by adding the robot URDF, defining foot contact points, building a joint mapping, correcting a left/right leg mismatch, and adding a virtual pelvis marker to reduce height mismatch between the human pose and the robot root.

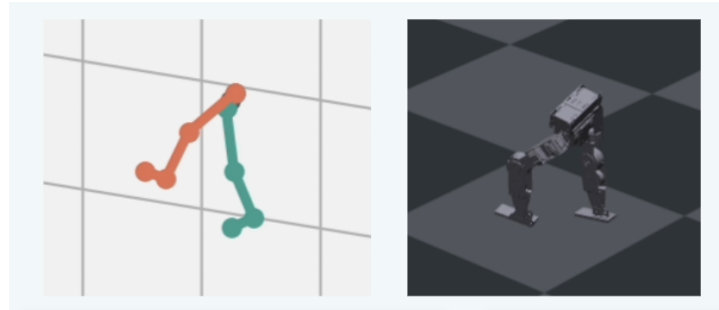


Fig. 4. Example human-to-robot retargeting result. Human lower-body pose estimates are mapped to the custom ch_robot model and visualized in simulation.

The hardest retargeting challenge was foot contact. Many human motions contain foot pitch or roll that the robot cannot reproduce while keeping stable contact. Hard flat-foot constraints sometimes improved foot orientation but caused infeasible frames or unnatural knee bending. Without those constraints, the robot feet

could remain inclined or slide. This mismatch motivated a simulation-first final MVP instead of direct real robot deployment.

3.7 Motion Tracking Policy and Simulation

The simulation stack evaluates whether the retargeted motion can drive a humanoid model in physics. During training, offline or recorded human motion is first retargeted into the robot joint format and then used as a reference motion. A policy is trained in IsaacLab using PPO. The final design uses residual PD control: the retargeted motion provides the nominal joint targets, and the policy outputs bounded corrective residuals around this reference:

$$q_j^{\text{des}} = \text{clip} \left(q_j^{\text{ref}} + s_{\text{res}} s_j a_{\pi,j}, q_j^{\text{min}}, q_j^{\text{max}} \right),$$

where q_j^{ref} is the retargeted reference joint position, $a_{\pi,j}$ is the policy action, $s_{\text{res}} = 0.15$ is the global residual action scale, and s_j is a joint-specific scale. The hip and knee residuals use larger correction ranges, while hip abduction/adduction and ankle joints are scaled down to reduce unstable corrections. This structure is more appropriate for motion tracking than directly commanding velocity because the policy only needs to stabilize and refine a desired motion rather than synthesize the full gait from scratch.

The reward was changed from velocity-command locomotion to reference motion tracking. Velocity-command rewards are useful when the robot only needs to walk at a desired speed, but teleoperation requires matching a specific reference pose sequence. The implemented reward uses exponential tracking terms for joint position, joint velocity, root planar position, root height, root yaw, root linear velocity, and yaw rate:

$$r_{\text{track}} = 4.0 \exp(-e_q/0.25) + 0.5 \exp(-e_{\dot{q}}/5.0) + 1.0 \exp(-e_{xy}/0.25) \\ + 1.0 \exp(-e_h/0.04) + 1.0 \exp(-e_{\psi}/0.25) + 0.5 \exp(-e_v/1.0) + 0.3 \exp(-e_{\dot{\psi}}/1.0),$$

where e_q is the mean squared actuated joint position error, $e_{\dot{q}}$ is the mean squared actuated joint velocity error, e_{xy} is the root planar position error, e_h is the root height error, e_{ψ} is the wrapped root yaw error, e_v is the body-frame root linear velocity error, and $e_{\dot{\psi}}$ is the yaw-rate error.

The full reward also includes an alive bonus and regularization penalties:

$$r = r_{\text{track}} + 0.05r_{\text{alive}} - 0.01c_{\Delta a} - 0.05c_{\Delta^2 a} - 0.0001c_{\dot{q}} \\ - 0.00001c_{\Delta \dot{q}} - 0.00002c_{\text{energy}} - 0.002c_{\text{overspeed}} - 0.10c_{\text{hip2}} \\ - 0.20c_{\text{limit}} - 0.05c_{\text{slip}} - 0.10c_{\text{undesired}}.$$

Here, $c_{\Delta a}$ penalizes action changes, $c_{\Delta^2 a}$ penalizes action jerk, $c_{\dot{q}}$ penalizes large joint velocities, $c_{\Delta \dot{q}}$ penalizes joint acceleration, c_{energy} penalizes mechanical power through $|\tau \dot{q}|$, $c_{\text{overspeed}}$ penalizes joints exceeding a soft velocity limit, c_{hip2} discourages excessive hip abduction/adduction, and c_{limit} penalizes joints near their position limits. Contact-based foot slip and undesired contact penalties are implemented in the reward, but they are only active when contact rewards and contact sensors are enabled. In the provided final configuration, these contact rewards are disabled, so the corresponding costs evaluate to zero during training. If the robot falls or violates the termination condition, the reward for that terminal step is replaced with a death cost of -1.0 .

The policy observation includes the robot's estimated up direction, root linear velocity, root angular velocity, actuated joint positions and velocities, previous actions, current reference tracking errors, reference joint velocities, root tracking errors, reference root velocity, yaw-rate information, and future reference joint errors. The single-frame observation has 108 dimensions and is stacked over three frames, giving the policy short-term temporal context while still keeping the controller reactive.

The final system uses IsaacLab for parallel PPO training and MuJoCo for additional replay and evaluation. This Sim2Sim path provides a safer intermediate validation step before real robot deployment. If a retargeted motion or policy fails across simulators, it is unlikely to be safe on the physical robot.

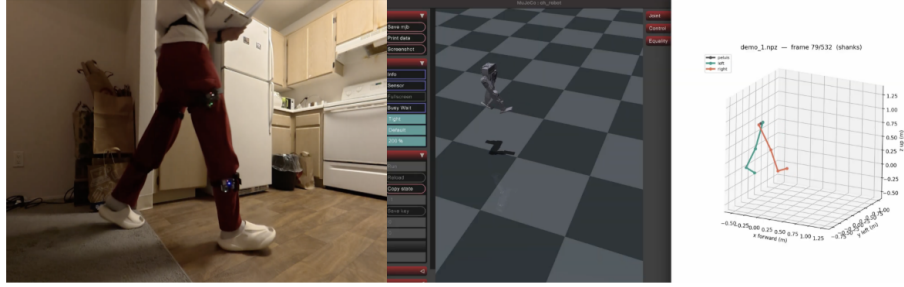


Fig. 5. End-to-end pipeline demonstration. A person wearing five IMUs on the lower body streams orientation data through the pose reconstruction and retargeting pipeline, driving the simulated `ch_robot` humanoid in real time. The pipeline covers wearable sensing, lower-body IK, human-to-robot retargeting, and simulation playback.

4 Milestones

4.1 Original Milestones

At the beginning of the quarter, the intended project pipeline was:

IMU wearables → human pose reconstruction → robot retargeting → motion tracking policy → MuJoCo evaluation → real human

The original plan included a seven-IMU lower-body setup, real-time teleoperation, retargeting to the custom humanoid model, reinforcement-learning-based motion tracking, and eventual testing on the physical robot.

4.2 Revised MVP

During the quarter, we revised the MVP toward simulation-based teleoperation. The revision had three main reasons. First, the five-IMU system was more reliable to demonstrate than the full seven-IMU suit, while still producing the nine keypoints required by the retargeting interface. Second, root translation and foot contact remained difficult with orientation-only sensing. Third, direct robot deployment would have been unsafe without more robust retargeting and policy validation.

The revised MVP became:

- build wearable IMU nodes and stream fused quaternions over UDP,
- reconstruct lower-body human pose from sparse IMU orientations,
- export pose clips in a standardized ML handoff format,
- retarget human lower-body motion to `ch_robot`,
- train and validate a motion-tracking policy in simulation,
- demonstrate live or recorded human motion controlling a simulated humanoid.

4.3 Completed and Deferred Milestones

Table 1 summarizes the final status of the major project milestones. Most of the core simulation-based teleoperation pipeline was completed, including wearable IMU sensing, multi-node wireless streaming, lower-body pose reconstruction, sparse keypoint generation, pose clip export, human-to-robot retargeting, and simulation-based motion tracking. These completed components form the final MVP because they connect the sensing layer to a reconstructed human pose, convert that pose into the robot representation, and validate the result in simulation.

Several goals were intentionally deferred. Real robot deployment was not attempted because the remaining issues involve safety-critical robustness, especially root translation, foot contact, retargeting feasibility, and policy

Table 1. Final project milestone status.

Milestone	Outcome	Status
Wearable IMU sensing	Built BNO085/ESP32-S3 nodes streaming 100 Hz fused quaternions over UDP.	Completed
Wireless communication	Validated packet reception from multiple nodes with sub-50 ms end-to-end latency.	Completed
Pose reconstruction	Implemented filtering, smoothing, neutral calibration, sensor alignment, and forward kinematics.	Completed
Sparse reconstruction	Produced nine lower-body retargeting keypoints from the five-IMU MVP setup.	Completed
Human pose handoff	Defined and implemented the .npz pose clip format for ML retargeting.	Completed
CV baseline	Built a RealSense/MediaPipe baseline for comparison and fallback sensing.	Completed
Retargeting	Adapted Holosoma to ch_robot and produced robot qpos trajectories.	Completed
RL motion tracking	Built IsaacLab PPO training with residual PD control and motion-tracking rewards.	Completed MVP
MuJoCo evaluation	Prepared simulation replay and IsaacLab-to-MuJoCo handoff assets.	Pipeline ready
Real robot deployment	Deferred because of retargeting robustness, foot contact, and safety concerns.	Future work
Full-body teleoperation	Deferred to focus on a lower-body MVP.	Future work

stability on hardware. Full-body teleoperation was also deferred so that the project could focus on a reliable lower-body MVP. This revision narrowed the scope of the project, but it made the final deliverable more realistic and better aligned with the technical progress completed during the quarter.

4.4 Problems Encountered and Management

The first major problem was embedded hardware reliability. Some BNO085 sensors occasionally booted into a non-responsive state. The firmware was modified to retry UART initialization and report enabling so that nodes could self-recover instead of requiring manual resets. This made the wearable nodes more reliable during demonstrations.

The second problem was wireless jitter. Early streams showed steady throughput interrupted by large stalls. We traced this to Wi-Fi power-saving behavior on the ESP32. Disabling Wi-Fi sleep and increasing transmit power made the 100 Hz UDP stream more stable.

The third problem was the limitation of orientation-only sensing. The IMU system reconstructs posture well, but it does not directly measure global pelvis translation. We managed this by supporting fixed-root reconstruction for posture and adding free-root foot anchoring for walking-like clips. However, root translation remains a limitation and should be improved with vision, external tracking, or stronger contact estimation.

The fourth problem was retargeting feasibility. Human and robot lower bodies do not have identical proportions, joint axes, or degrees of freedom. Foot contact was especially difficult: hard flat-foot constraints sometimes produced unnatural knee configurations, while softer constraints could leave the feet inclined. This led us to defer real robot deployment and validate the system in simulation first.

The fifth problem was reinforcement-learning reward design and compute availability. Velocity-command rewards were not sufficient for motion tracking, so we rewrote the policy setup around residual PD control and reference-motion tracking. Near the end of the quarter, GPU access limited the amount of training and tuning that could be completed.

5 Conclusion

This project built a low-cost lower-body humanoid teleoperation pipeline using wearable IMUs, ESP32-S3 wireless nodes, human pose reconstruction, retargeting, reinforcement learning, and simulation validation. The final MVP connects live or recorded IMU sensing to a reconstructed human lower-body pose, exports that pose in a standardized ML handoff format, retargets it to a custom humanoid robot model, and evaluates the result in simulation.

The significance of the project is practical: it replaces expensive motion capture hardware with a lower-cost wearable sensing system while preserving a path toward real humanoid teleoperation. The project also clarifies the main technical challenges that must be solved before real robot deployment, including robust calibration, root translation estimation, contact-aware retargeting, policy robustness to noisy input, and Sim2Real safety validation.

Future work should improve functional calibration, add magnetometer disturbance handling, integrate vision or another source of pelvis translation, train on larger locomotion datasets, and validate policies across simulators before controlled real robot deployment. Full-body teleoperation is also a natural extension once the lower-body pipeline is stable.

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